

Yashika Basapure

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SUMMARY

An Engineer with experience in deep learning, Bayesian inference, geospatial pipelines, and production ML systems. Hands-on with LLMs, LVMs, satellite imagery, and end-to-end data engineering.

EDUCATION

JSPM, Pune, India <i>B.E. in Computer Science</i> CGPA: 8.7/10	Pune, India <i>July 2026</i>
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TECHNICAL SKILLS

Languages: Python, Java, C++, SQL
ML & AI: TensorFlow, PyTorch, Speech Recognition (ASR), R-CNN, HuggingFace, Transformers, LLMs, LVMs/VLMs, Deep Learning
Geospatial & Data: Satellite Imagery, Remote Sensing, Raster & Vector Data, PostGIS, PostgreSQL, MongoDB, Pandas, NumPy
Web & Backend: React, Redux, Flask, FastAPI, REST APIs, Postman, HTML/CSS
Cloud & DevOps: Microsoft Azure, Virtual Machines, Docker, Git, PaaS, AIaaS
Soft Skills: Leadership, Problem Solving, Team Collaboration, Communication

INTERNSHIP EXPERIENCE

Data Scientist Jr. <i>Agtmap.com</i>	Remote <i>Jan 2026 – April 2026</i>
- Built end-to-end geospatial data pipelines for satellite imagery, raster, and vector datasets including preprocessing, geo referencing, and transformation with a focus on data quality at scale. - Applied LLMs and LVMs to multimodal geospatial tasks, including segmentation using SAMGeo and basic evaluation of satellite data; deployed ML pipelines on Microsoft Azure for scalable training and inference. - Integrated and managed geospatial data (GeoAI features, raster, vector) in PostGIS/PostgreSQL; worked with Virtual Machines and cloud environments for model experimentation and data processing.	

RESEARCH EXPERIENCE

Virtual Brain Inference Research <i>Independent Research Project</i>	Pune, India <i>2025</i>
- Engineered modular VBI pipeline integrating diffusion MRI connectomes with neural mass models (Jansen-Rit, Wong-Wang, Epileptor-lite) for GPU-accelerated brain simulation. - Implemented Bayesian inference for personalized parameter estimation enabling uncertainty-quantified virtual brain twins for clinical seizure prediction.	

PROJECTS

Fraud Detection in Digital Payment Systems <i>Python, XGBoost, SMOTE, SHAP</i>	2025
- Built a robust fraud detection system processing 100K+ financial transactions, training and benchmarking Logistic Regression, Random Forest, and XGBoost models with systematic hyperparameter tuning to maximize detection performance on a highly imbalanced dataset. - Applied SMOTE oversampling combined with cost-sensitive learning to address class imbalance, resulting in an 18% improvement in fraud recall while maintaining precision — directly reducing false negatives that carry high financial risk. - Integrated SHAP and LIME explainability frameworks to generate feature-level prediction explanations, enabling compliance with financial regulatory standards and providing actionable insights for fraud analysts.	
Hindi ASR Fine-tuning — Whisper on Conversational Hindi <i>Transformers, Whisper, Librosa, PyTorch</i>	2026
- Fine-tuned OpenAI Whisper-small on 10hrs of conversational Hindi audio (4,929 segments across 104 speakers) — reduced Word Error Rate from 178% to 12.25% through domain-specific training. - Built end-to-end data pipeline: automated GCS audio download, timestamp-based segmentation, 16kHz resampling, and train/test splitting using Librosa and SoundFile. - Implemented ASR cleanup pipeline Hindi number normalization and Roman-script English word detection/tagging for downstream NLP tasks. - Designed lattice-based WER evaluation system to fairly score ASR models against multiple valid transcription alternatives — reducing unfair penalisation by 100% on number variant errors. - Applied text normalisation post-processing fix, further reduced WER from 12.25% to 10.57%.	

ACHIEVEMENTS

C++ Gold Badge, HackerRank (2025) Finalist, SmartAI Hackathon (2024) — ML fraud detection system Top 3, Data Nexus Hackathon (2024) — React browser extension with live traffic integration
